METHOD FOR GENERATING A CHANNEL-OPTIMIZED DATA ENCODING MAP

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Abstract

A method for generating a data encoding map mathematically optimized for a high-cardinality parallel communication channel. The method comprises: determining the parameters of a communication channel, including the number of parallel channels (M) and the number of unique states each channel can represent (N); defining a data block size (B) in bytes or bits; and computationally generating a one-to-one mapping between every possible data permutation within the block size (B) and a unique state (from N states) of a single channel (from M channels). This method creates a non-human-readable, machine-optimized "character set" that enables the transmission of multi-byte data blocks within a single state of a single channel, achieving maximum possible data density for the given physical layer.

Background of the Invention

Traditional data encoding methods, such as ASCII and UTF-8, were designed for human-readable text and were predicated on the physical limitations of communication channels at the time: serial transmission of binary data (1s and 0s). The channel itself had a very low cardinality (two states). The advent of new physical layers, particularly massively parallel optical interconnects, has created channels with extremely high cardinality, where a single channel (e.g., a pixel) can represent millions of unique states. Legacy encoding methods are inefficient for such channels, as they still require transmitting sequences of binary bits, failing to utilize the full state-space of the channel. A new method is needed to create encoding maps designed for these new high-cardinality channels.

Summary of the Invention

The present invention is a method for generating a channel-optimized data encoding map that is tailored to the specific capabilities of a high-cardinality, parallel communication interface. Instead of using a pre-defined, human-readable character set, this method creates a dense, machine-optimized encoding for maximum data throughput.

The method begins by analyzing the physical communication channel to determine its parameters: M, the number of parallel channels (e.g., pixels in an array), and N, the number of distinct states each channel can represent (e.g., $2\hat{2}4$ for a 24-bit color pixel). A data block size, B, is then chosen such that the number of possible data combinations in the block (e.g., $2\hat{(}8*B)$ for B bytes) is less than or equal to N. The core of the method is a computational process that generates a map, or lookup table, which creates a unique, one-to-one correspondence between every possible data pattern of size B and a unique state of the N possible channel states.

This generated map is then used by the communication endpoints to encode and decode data. For example, a 3-byte block of data can be mapped directly to a single 24-bit pixel color, transmitted, and then decoded by the receiver using the same map. This method enables a level of data density that is impossible with traditional encoding schemes.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of the method for generating the channel-optimized encoding map.

Detailed Description of the Invention

Referring to **FIG. 1**, the method for generating a channel-optimized data encoding map is shown. The process begins with analyzing the physical communication channel to determine its parameters (101). These parameters are the number of parallel channels, M, and the number of unique states per channel, N. Next, a data block size, B, is selected (102) such that the total number of possible data permutations of size B is less than or equal to N.

A computational process is then initiated to generate the encoding map (103). This process iterates through every possible unique data block of size B and assigns it a unique channel state from the N available states, creating a one-to-one mapping. The result is a complete, bijective map (104) where any given data block corresponds to exactly one channel state, and vice versa. This map is then distributed to the communication endpoints (105) to be used for encoding and decoding data during a communication session.

Claims

What is claimed is:

1. A method for generating a data encoding map for a communication channel, the method comprising the steps of:

- determining that a communication channel comprises a plurality of parallel channels, wherein each channel is capable of representing N unique states, where N is greater than 2;
- selecting a data block size, B, such that the number of unique data permutations of size B is less than or equal to N; and
- computationally generating a map that provides a one-to-one correspondence between each unique data permutation of size B and a unique state of the N states.

2. The method of claim 1, wherein the parallel channels are pixels in a two-dimensional array.

3. The method of claim 2, wherein the N unique states are the possible color values of a pixel.

4. The method of claim 3, wherein N is $2\hat{2}4$ and the data block size B is 3 bytes.

5. The method of claim 1, further comprising the step of transmitting the generated map to at least two communication endpoints for use in encoding and decoding data.

6. The method of claim 1, wherein the communication channel is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.

Drawings

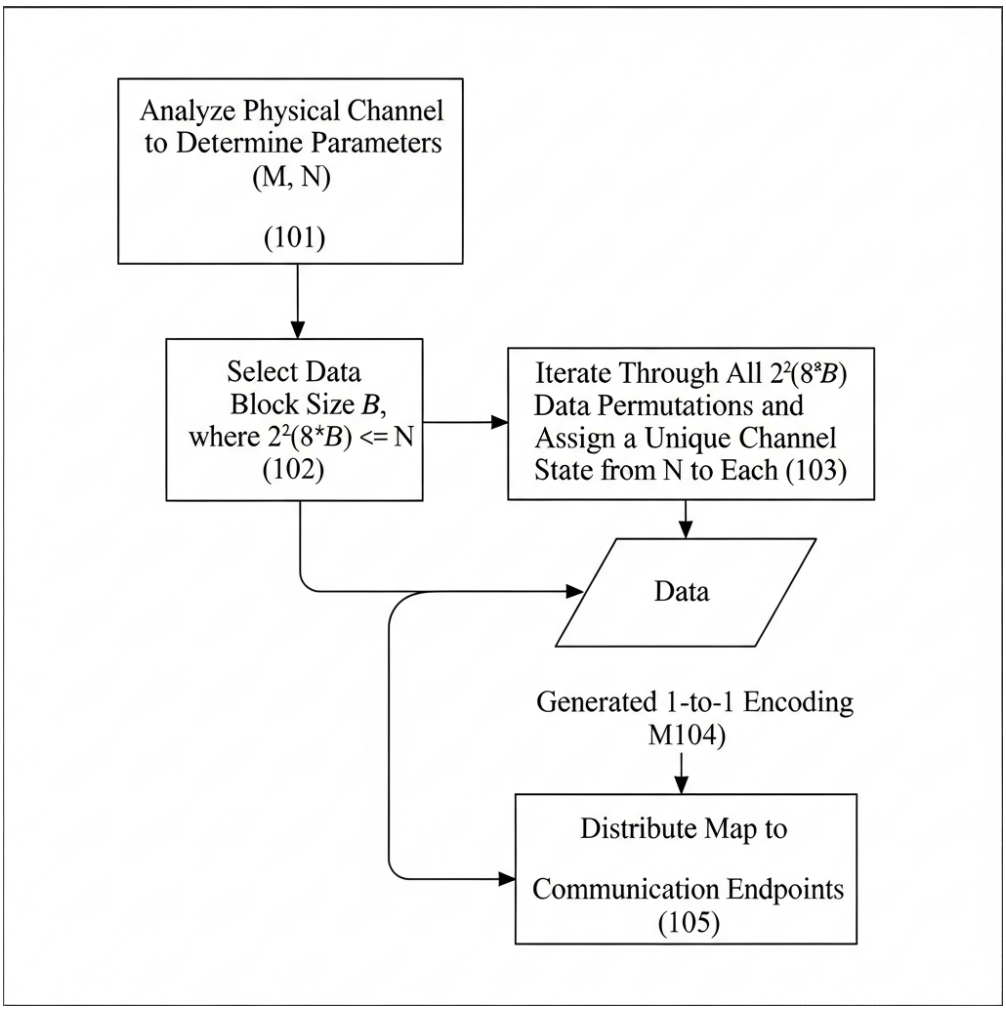


Figure 1: Flowchart illustrating the steps of the method for generating the channel-optimized encoding map.